



BITS Pilani
K K Birla Goa Campus

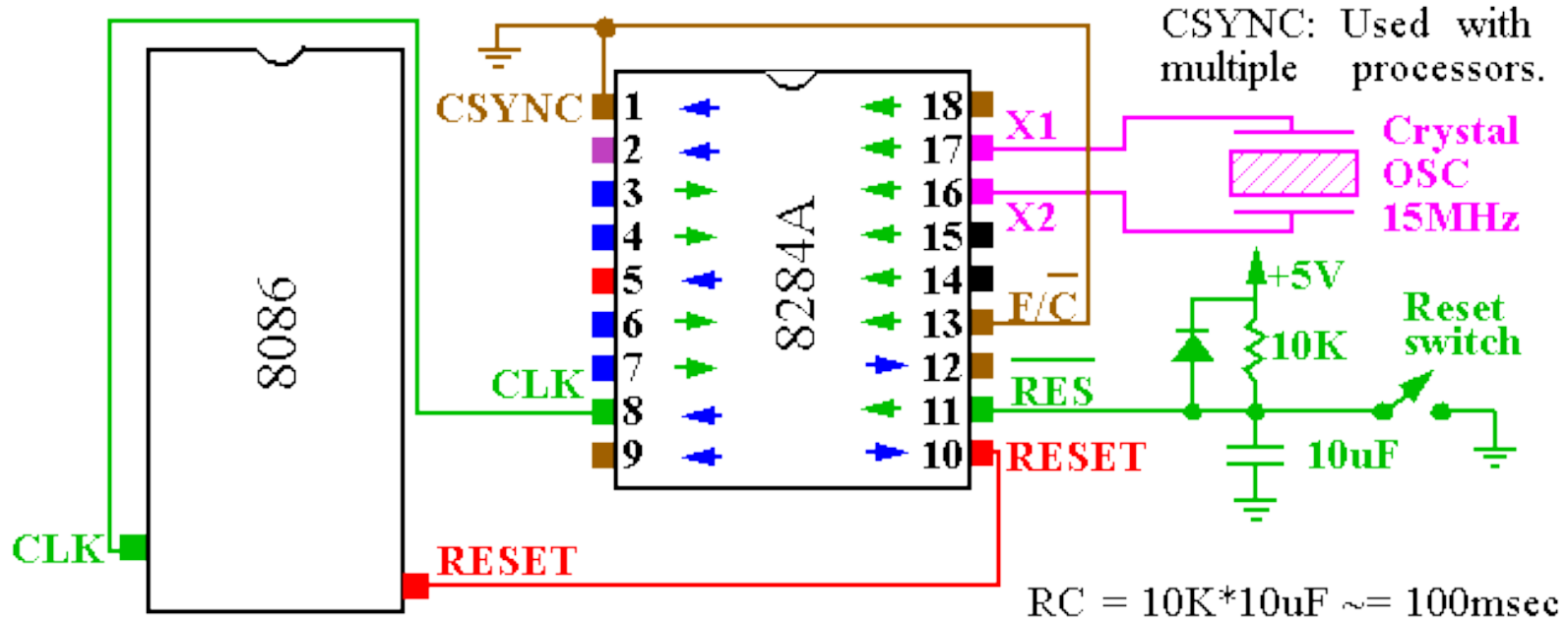
8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

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Department of CS & IS



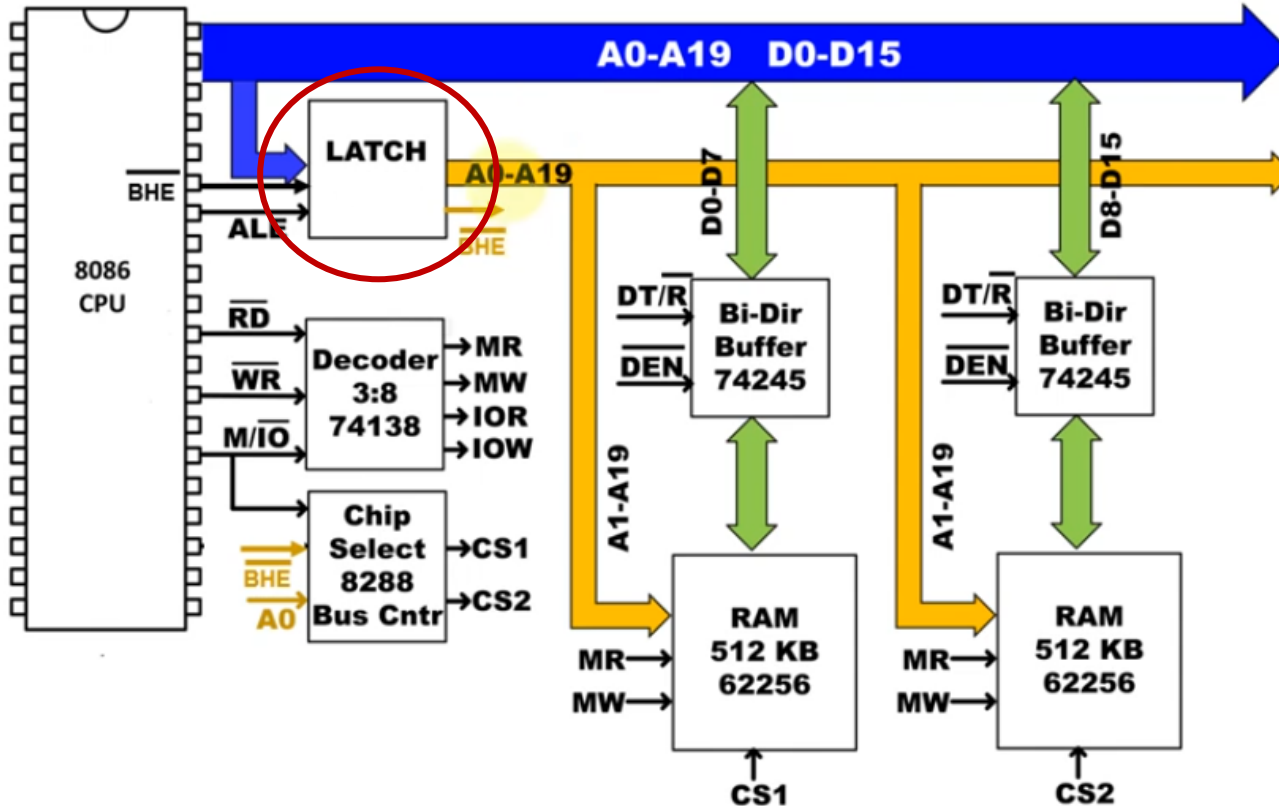
Reset Operation RC Circuit



Microprocessor resets if this RESET pin is held high for 4 clock periods. Instruction execution begins at FFFF0H and IF flag is cleared.

Overview

- Bus Demultiplexing
- Bus buffering
- Timing Diagram



BUS Demultiplexing

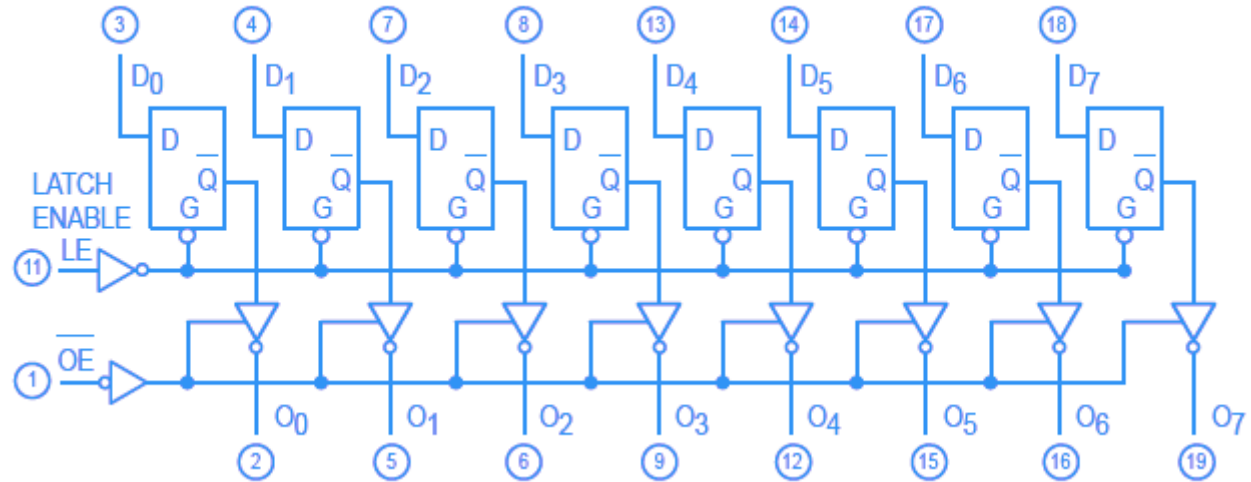
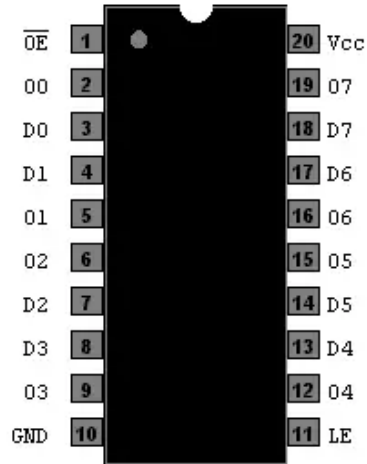
Demultiplexing the Buses:

Computer systems have three buses:

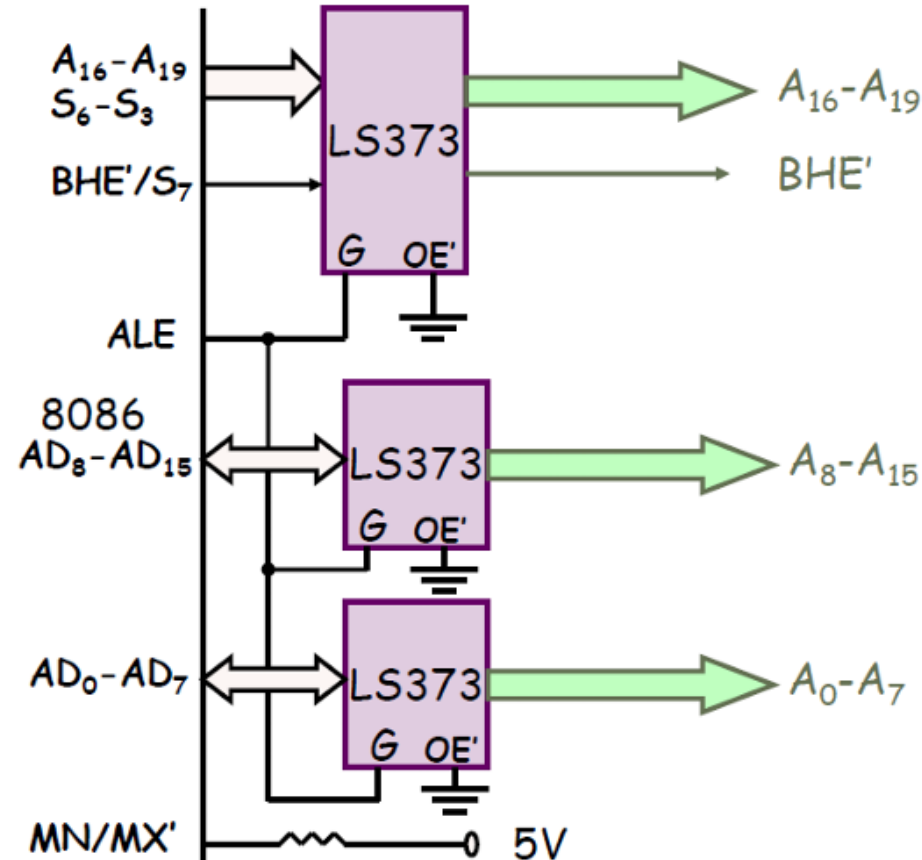
- Address (20 bit)
- Data (8 bit for 8088, 16 bit for 8086)
- Control

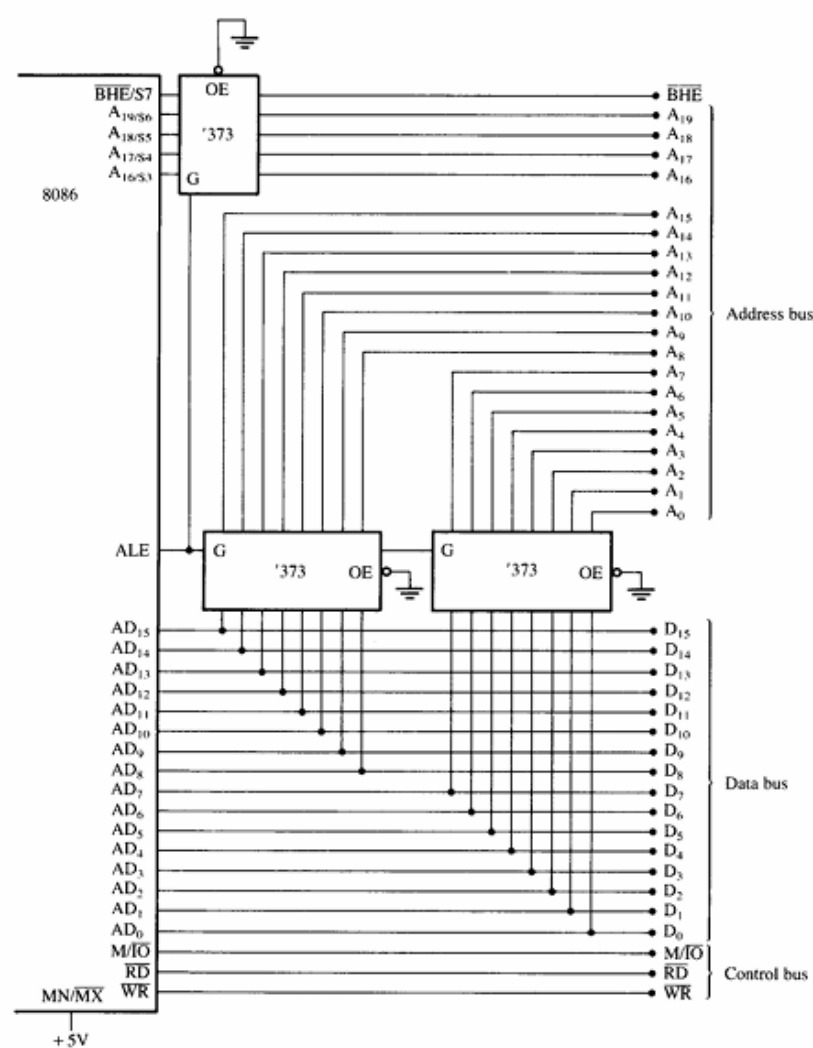
The Address and Data bus are multiplexed (shared) due to pin limitations on the 8086.

Latch IC 74LS373

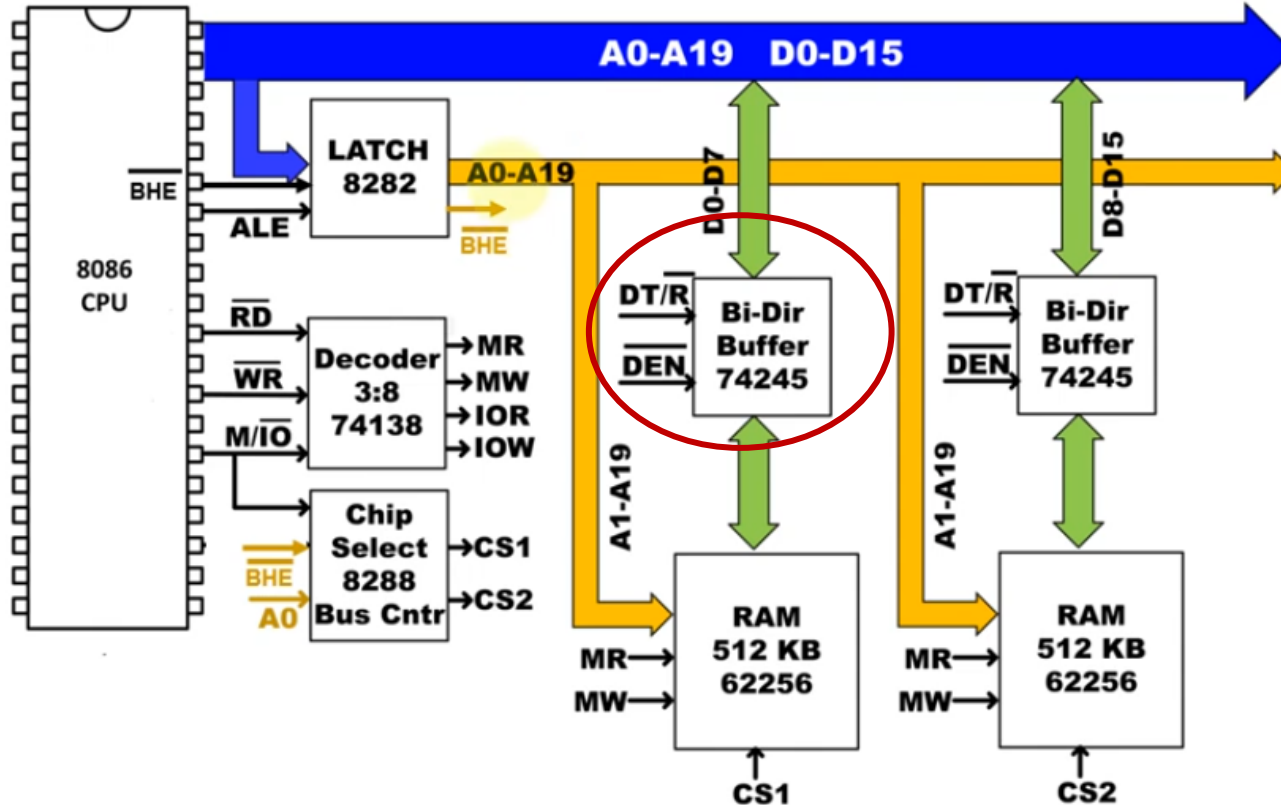


BUS De-Multiplexing

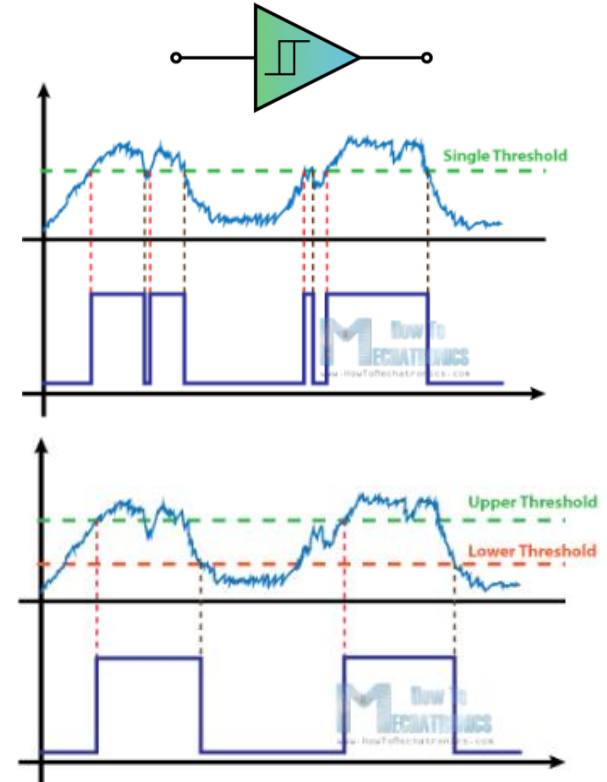
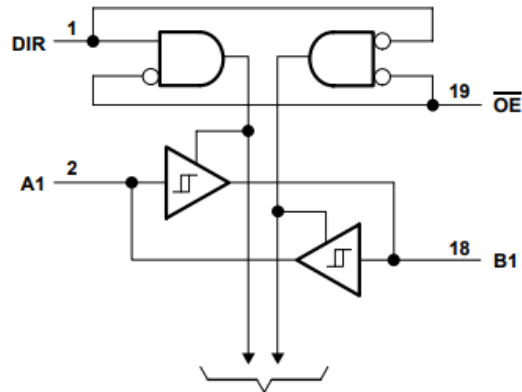
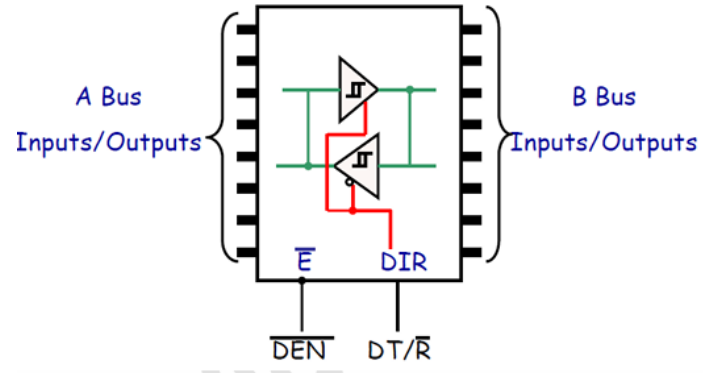


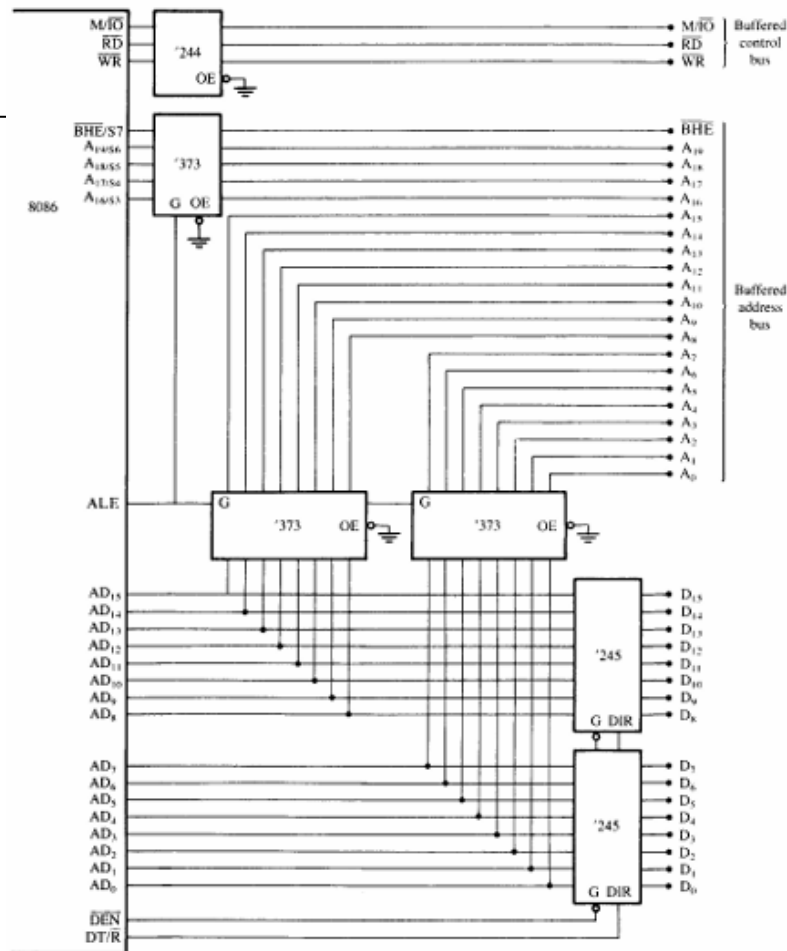


Bus Buffering

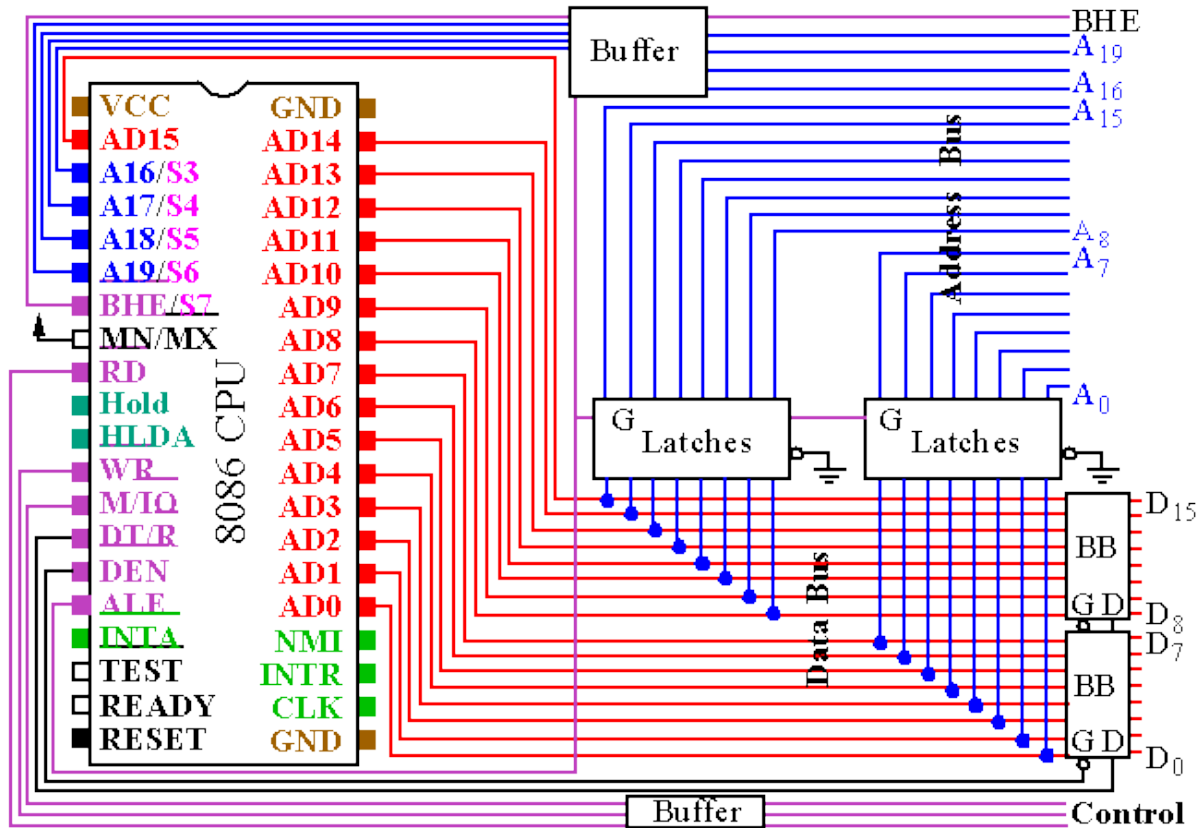


74LS245 Buffer

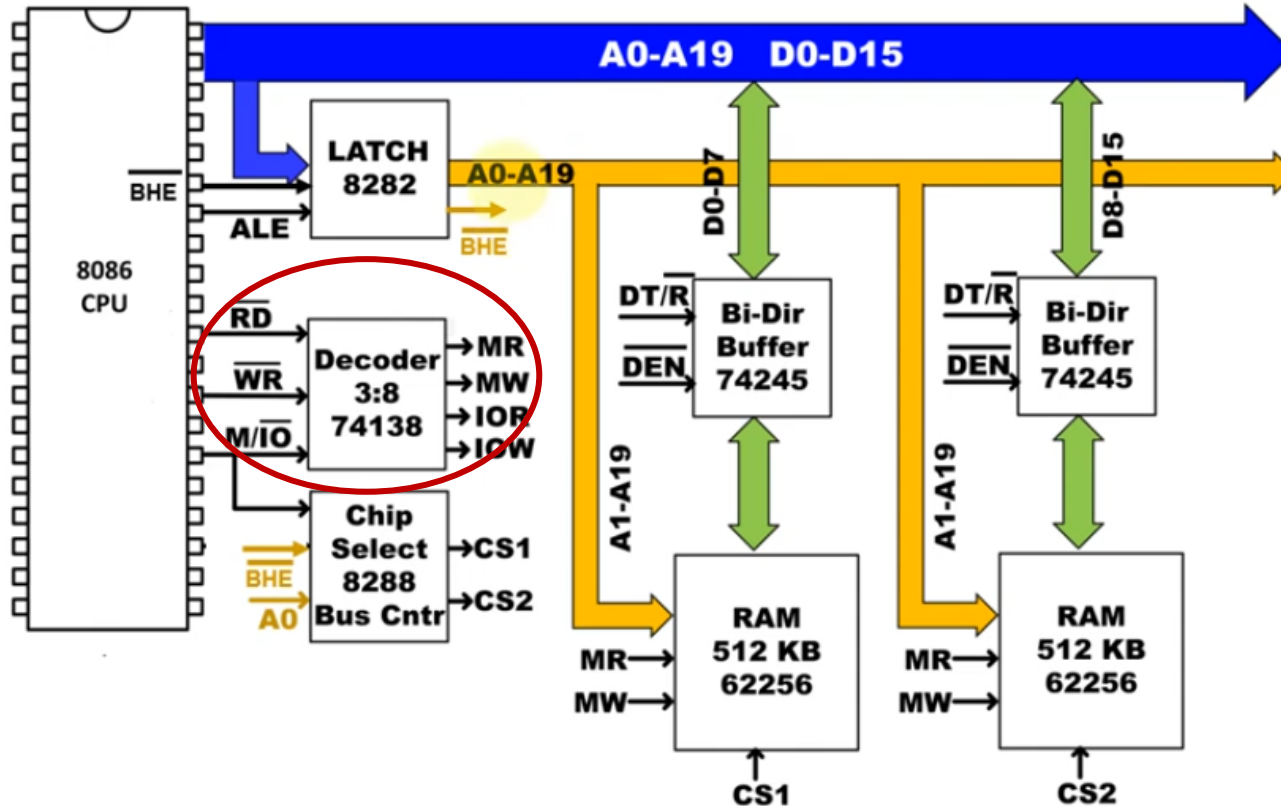




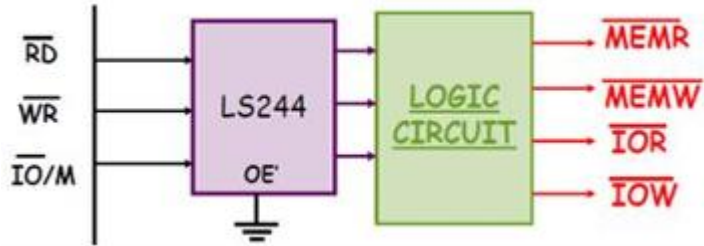
8086 Buffered and Demultiplexed buses



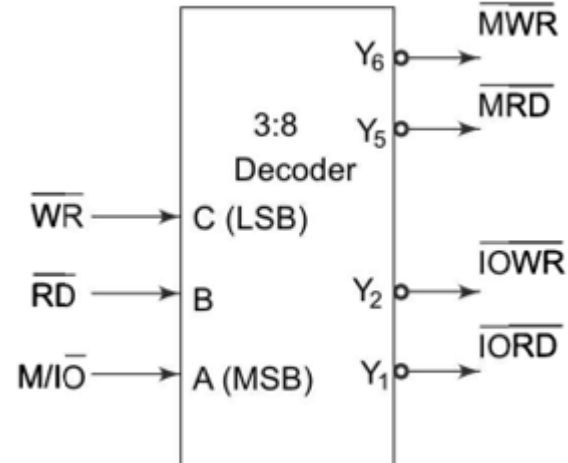
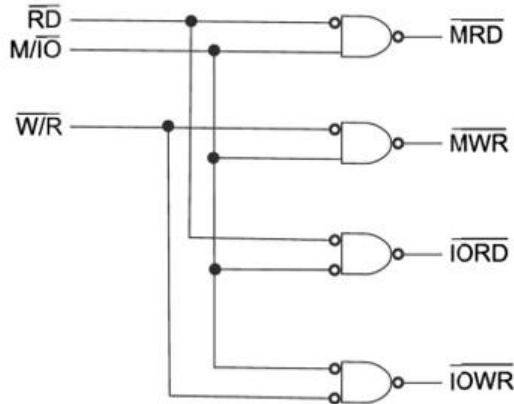
Decoder



Decoder



M/IO'	RD'	WR'	FUNCTION
1	0	1	MEMR
1	1	0	MEMW
0	0	1	IOR
0	1	0	IOW

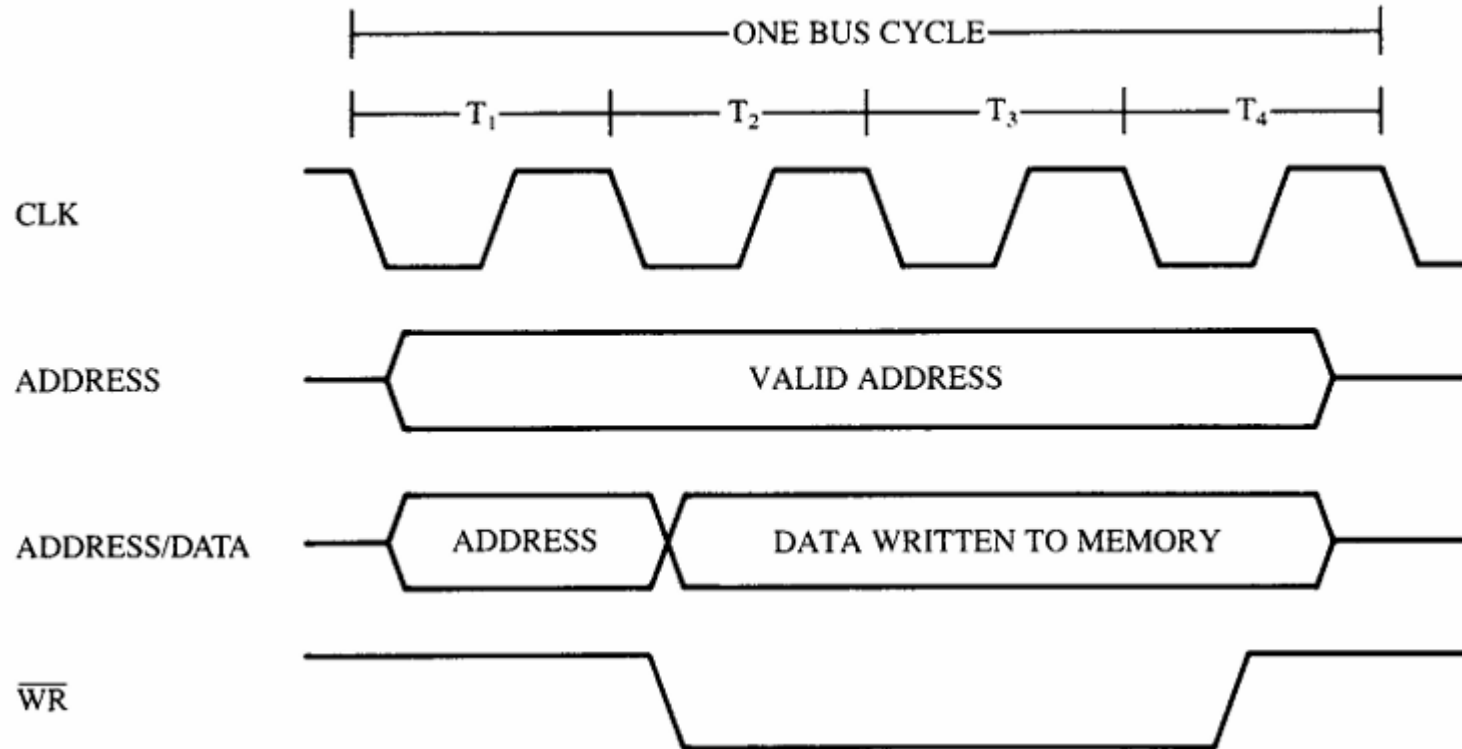


Bus Timing

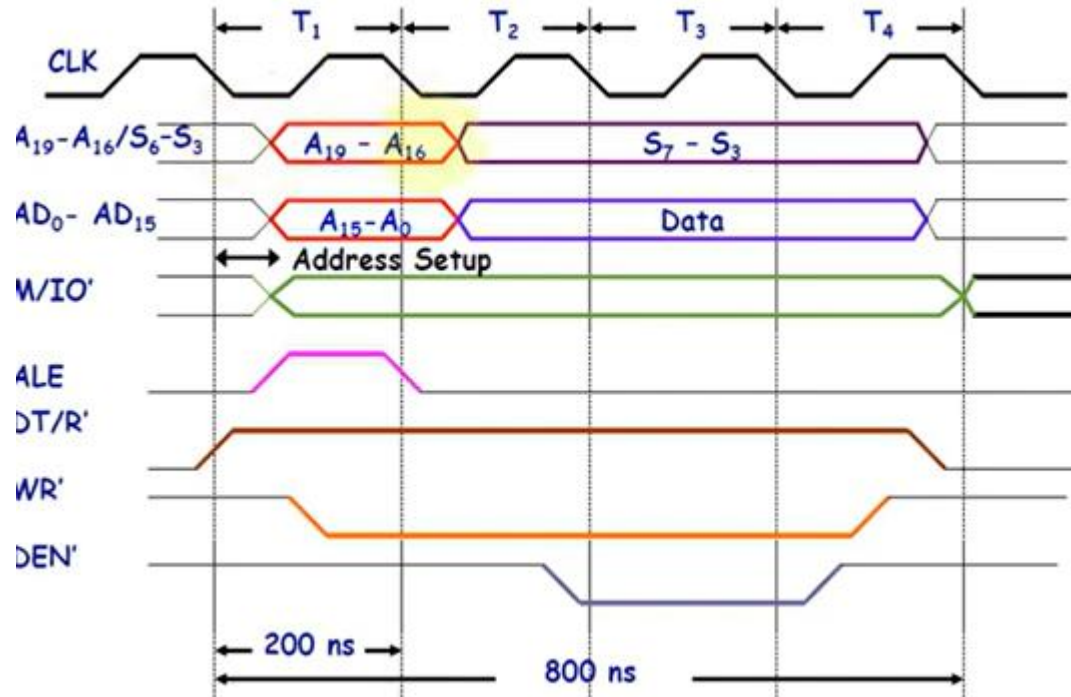
BUS cycle /Machine Cycle: Time used to access the memory, peripheral devices (I/O devices) and interrupt controller. It corresponds to sequence of events starting with address being placed on system address bus followed by read or write data transfer.

Each bus cycle equals to **four** system-clocking period (T states).

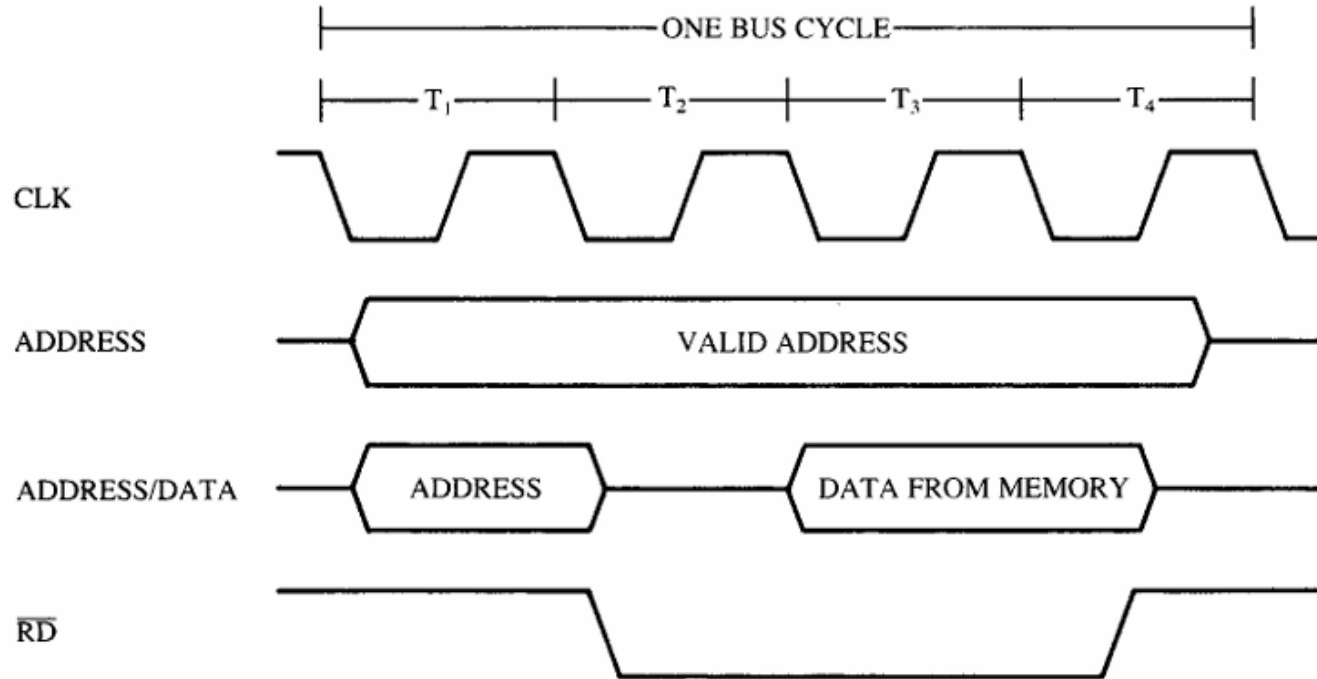
Timing Diagram Write operation



Memory Write Operation



Timing Diagram Read operation



Memory Read Operation

